High-Precision BLDC Motor Control Using a Custom ESP32-S3 PCB

Computer Architecture Project Group
University of Basel, Fall Semester 2026

Working report – design and evaluation scaffold

Draft status. This document is a writing scaffold. Hardware, firmware, and experiments have not yet been implemented or evaluated. Replace the prompts below with documented work and measured results.

1 Purpose and approach

The project aims to develop a custom ESP32-S3-WROOM-based PCB for closed-loop angular positioning of a brushless DC motor, with rotor-position feedback and field-oriented control as the intended strategy.

The provisional target is a steady-state absolute angular error of at most 1° under no external load, subject to motor and sensor selection. This is not a measured result. Off-the-shelf controller boards are the planned fallback if the custom PCB cannot be commissioned.

To complete: explain the application, agreed requirements, system architecture, and relevant computer architecture concepts.

2 Hardware design

To complete: justify motor, sensor, driver, and power-supply selection; describe the schematic, PCB, electrical interfaces, BOM, assembly, and commissioning. Identify the board revision and link the design files.

3 Firmware and control

To complete: describe the toolchain, control strategy, timing, peripheral use, calibration, operating limits, and build/flash procedure. Reference the evaluated firmware commit.

4 Experimental evaluation

To complete: define the measurement reference, uncertainty, load, calibration, and test procedure. Report positioning error, repeatability, stability, and step response using measured data. Compare results with agreed targets and discuss limitations.

5 Contributions and conclusions

To complete: record each member's work, summarize achieved results, explain difficulties and design changes, and identify further improvements. Everyone should be able to explain their contribution in the final presentation.